

Research Assignment Module 1

1. Business Analytics

- It's the use of data, statistical methods, models and analytical techniques to understand business performance and support decision making. (IBM, 2023)
e.g Investment maturities analysis

⇒ Distinction: Business analytics focuses on using analysis to solve business problems, while data analysis focuses more on analysing the data itself.

2. Data Analytics

- It's the process of collecting, cleaning, transforming and analysing data to identify patterns, trends and useful information. (IBM, 2025)
e.g Analysing fx data trends.

⇒ Distinction:

Data Analytics → What does the data tell us.

Business Analytics → How can we use that information to make a better business decision.

3. Business Intelligence

- It refers to the systems used to transform organisation data into accessible information for monitoring performance and decisions. (IBM, 2021).

e.g. Power BI dashboards

- Distinction $\Rightarrow$ BI focuses on reporting and monitoring what has happened or is happening, while BA can go further by using statistical techniques to determine what may happen, and what should be done.

4. Data driven decision making

- It means making decisions primarily on data and evidence rather than relying on intuition. (Tableau, 2026)
e.g. compare quoted rates before making an investment decision.

- The important point is that analytics produces information, data-driven decision making is the practice of actually using that information to make decisions.

5. Big data

- Refers to datasets that are so large, fast moving that traditional data processing may struggle to handle them efficiently. (Oracle, 2024).
- Characteristics: Volume, Velocity, Variety.
- Distinction $\Rightarrow$ Big data does not simply mean a large excel sheet. It involves scale, speed requiring more advanced data infrastructure.

6. Descriptive Analytics

- It summarises historical data to explain what has already happened
- techniques include totals, averages.

7. Diagnostic Analytics

- Goes beyond describing an outcome and investigates why it occurred (IBM, 2025).
- Distinction
 - Descriptive = identifies outcome
 - Diagnostic = investigates causes

8. Predictive Analytics

- Uses historical data to estimate future outcomes, (IBM, 2024).
- Methods include regression, time series models.

9. Prescriptive Analytics

- Uses data, forecasts and analytical models to determine or compare possible outcomes. (IBM, 2024)

10. Data Literacy

- It is the ability to understand, interpret, analyse, and communicate data effectively

02. Data, Measurement & Reporting

11. Structured Data

- It is information organised according to a predefined format, usually in rows and columns. (IBM, 2005)
e.g. A treasury spreadsheet containing columns for transaction dates.

12. Unstructured Data

- It's information that does not follow a predefined structure or fixed format. (IBM, 2005)
e.g. Customer email.

13. Data Warehouse

- It's a centralised repository that combines data from different sources and stores it for reporting, analysis and business decision-making. (Oracle, 2003)
e.g. Combined historical cash-flow.

14. Data Mining

- It's the process of examining large amounts of data to discover useful patterns, relationships and trends that may not be immediately obvious. (IBM, 2005)
e.g. A bank could analyse historical loan data.

15. Data Visualization

- Means ~~visualize~~ presenting data visually through charts, graphs, maps, tables and other graphic formats.
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e.g. A line graph displaying monthly investment return.

16. Dashboard

- A visual interface that brings together important metrics, charts and performance information in one place.
e.g. A treasury dashboard displaying available cash.

17. Metric

- Any measurable value used to track or describe an activity or performance. (
e.g. A company processed 1200 transactions during September.

18. Key Performance Indicator (KPI)

- A metric specifically selected to measure progress towards an important business objective or target.
(Microsoft, 2026).

19. Benchmark

- A reference point or standard used to compare performance
e.g. Using Jibar/Zarona to compare with interest rates.

20. Segmentation

- Process of dividing a larger group into smaller groups based on shared characteristics or attributes.
e.g. A lender could segment customers according to loan sizes.

2.3 Customer & sales KPIs

21. Conversion Rate

- Measures the percentage of people who complete a desired action out of the total number of people who had the opportunity to do so. (Google, 2026)

Formula:

$$\text{Conversion rate} = \left(\frac{\text{No. of Conversions}}{\text{Total Visits/Leads}} \right) \times 100$$

22. Customer Acquisition Cost (CAC)

- Measures how much a business spends, on average, to acquire one new customer (Qualtrics, 2026).

$$\text{CAC} = \frac{\text{Total sales and Marketing cost}}{\text{Number of New Customers Acquired}}$$

23. Customer Lifetime Value (CLV)

- Estimated economic value a business expects to earn from a customer over the entire customer relationship. (Qualtrics, 2026).

$$CLV = \text{Average Customer Profit per Period} \times \text{Average Customer Lifespan.}$$

24. Churn Rate

- Measures the percentage of customers who stop using a company's product or service during a particular period (Qualtrics, 2026).

$$\text{Churn Rate} = \left(\frac{\text{Customer Lost During Period}}{\text{Customers at start Period}} \right) \times 100$$

25. Customer Retention Rate

- Measures the percentage of existing customers that a company retains over a particular period

$$\text{Retention Rate} = \left(\frac{\text{Customer at end} - \text{New Customer Acquired}}{\text{Customer at start}} \right) \times 100$$

26. Net Promoter Score (NPS)

- Measures customer's willingness to recommend a company, product or service to others (Qualtrics, 2026).

$$NPS = \% \text{ Promoters} - \% \text{ Detractors}$$

27. Average Order Value (AOV)

- Measures the average amount of revenue generated from each customer order or transaction. (Shopify, 2026).

$$\text{AOV} = \frac{\text{Total Revenue}}{\text{No. of Orders}}$$

28. Sales funnel

- Represents the stages potential customers move through before making a purchase (Salesforce, 2026).

Funnel: Awareness → Interest → Consideration →
Conversion / Purchase

04. Marketing & Campaign Analytics

29. Marketing Campaign

- This is a coordinated set of activities designed to achieve a specific objective (Hubspot, 2026).
e.g. Increasing product sales

30. Post Campaign Analysis

- It's the evaluation of a marketing campaign after it has ended to determine how well it performed against its objectives (Hubspot, 2026).
e.g. Increase in revenue.

31. Incremental Revenue

- Refers to the additional revenue generated as a result of a particular marketing activity or campaign, compared with the revenue that would have been generated without it. (Google, 2026)

32. Return on Investment (ROI)

- The profitability of an investment by comparing the financial return generated with its cost. (Mailchimp, 2026).

$$ROI = \frac{(\text{Return} - \text{Cost})}{\text{Cost} \times 100}$$

33. Return on Ad Spend (ROAS)

- Measures the revenue generated for every unit of money spent specifically on advertising (Mailchimp, 2026)

$$ROAS = \frac{\text{Revenue attributed to Advertising}}{\text{Advertising Cost}}$$

34. Click Through Rate (CTR)

- Measures how often people who see an advertisement or link click on it (Google, 2026)

$$CTR = \frac{\text{Clicks}}{\text{Impressions}} \times 100$$

35. Bounce Rate

- Measures the percentage of website sessions that are not considered engaged sessions (Google, 2026)

36. A/B Testing

- Method of comparing two versions of a marketing element, such as an advertisement to determine which performs better (Optimizely, 2026).

37. Lead Generation

- The process of attracting potential customers and encouraging them to provide information or show interest in a company's products or services (Hubspot, 2026).

38. Marketing Attribution

- The process of determining which marketing interaction or channels contributed to a desired outcome, (Hubspot, 2026)

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